

Retail Sales & Business Performance Analysis

Beginner-Level Data Analyst Internship Project

1. Introduction

This report documents a Beginner-Level Data Analyst internship project that analyzes a retail/e-commerce sales dataset and converts raw transactional data into KPIs, trend analysis, visualizations, a spreadsheet dashboard, and business recommendations.

2. Business Problem

Main question: How is the retail business performing, and which products, categories, and regions are driving revenue and profit — and where are the opportunities for improvement?

Thirteen supporting questions guided the analysis, covering total revenue and profit, order and customer counts, average order value, monthly performance, category and product performance, regional performance, and customer contribution. The full list is in `documentation/methodology.md`.

3. Dataset Description

A synthetic retail dataset was generated to mirror a realistic e-commerce export — this is explicitly disclosed, as no real transactional data was used. It spans **January 2024 to March 2025** (15 months), with 2,233 raw transaction records across 5 categories (Electronics, Furniture, Clothing, Home & Kitchen, Stationery), 4 regions (North, South, East, West), and 386 unique customers.

4. Data Cleaning

The raw export contained realistic data-quality issues: duplicate rows, missing values, inconsistent category/region spelling, invalid dates, and a small number of invalid numeric entries. Each issue was inspected, documented, and resolved:

- **25** exact duplicate rows removed
- **11** rows with invalid/unparseable dates removed
- **8** rows with invalid numeric values (Quantity or Unit Price ≤ 0) removed
- Category and Region spelling/casing standardized
- **44** missing Customer Name values filled with “Unknown Customer” (row kept — still a valid transaction)
- **66** missing Discount values filled with 0% (no discount recorded = no discount applied)

Final cleaned dataset: **2,189 records**. Full documentation: `documentation/data_cleaning.md`.

5. Methodology

Workflow: Business Problem → Dataset → Inspection → Cleaning → Transformation → KPIs → Monthly/Category/Product/Regional/Customer/ Profitability Analysis → Dashboard → Insights → Recommendations → Report.

Derived columns added during transformation: Year, Month, Month Number, Year-Month, Profit Margin %, Order Value.

All KPIs and analysis tables were built as live Excel formulas (SUMIFS, COUNTIFS, SUMPRODUCT, INDEX/MATCH) referencing the Cleaned Data sheet directly, so the workbook recalculates automatically if the underlying data changes.

6. KPI Definitions

KPI	Definition	Value
Total Sales	Sum of all Sales	₹91,12,531
Total Profit	Sum of all Profit	₹22,05,548
Total Orders	Count of Order IDs	2,189
Total Quantity Sold	Sum of Quantity	3,029
Total Customers	Unique Customer IDs	386
Average Order Value	Total Sales ÷ Total Orders	₹4,163
Profit Margin %	Total Profit ÷ Total Sales	24.2%
Top Category (Sales)	—	Electronics
Top Product (Sales)	—	Bluetooth Speaker
Best Region (Sales)	—	North

7. Analysis

Monthly: October 2024 and November 2024 were the two strongest months (₹8.37L and ₹7.43L in sales); January was consistently the weakest month in both years, consistent with a festive-season demand pattern.

Category: Electronics led in sales (₹27.68L) but had the lowest margin (10.1%). Stationery had the smallest sales base (₹9.07L) but the highest margin (48.6%) — a clear demonstration that sales volume and profitability do not move together.

Product: Bluetooth Speaker was the top seller by revenue but returned a below-average 9.9% margin. USB-C Charger was weak on both sales and margin (3.2%) — the only product weak on both dimensions simultaneously.

Regional: All four regions performed within a narrow band; North led narrowly on both sales and margin. East had a smaller customer base rather than a profitability problem.

Customer: 314 of 386 customers (81%) were repeat buyers. Revenue was not concentrated — the top customer contributed under 2% of total sales.

Profitability: Discount depth was directly correlated with declining margin, turning negative (-0.6%) at the 30% discount tier.

8. Dashboard Explanation

The workbook's Dashboard sheet (first tab) presents six KPI cards (Total Sales, Total Profit, Total Orders, Total Customers, Average Order Value, Profit Margin %), three dropdown filters (Year, Region, Category) that recompute the KPI cards live, and four charts: a monthly Sales & Profit trend line, a Category Sales & Profit comparison, a Regional Sales comparison, and a Top 10 Products by Sales bar chart.

9. Key Findings

1. Electronics drives the most revenue but returns the lowest margin of any category.
2. High sales does not equal high profit — confirmed directly by category ranking.
3. October–November is a clear seasonal sales peak.
4. Discounting above 20% consistently erodes profit margin; 30% is unprofitable on average.
5. USB-C Charger underperforms on both sales and margin.
6. Bluetooth Speaker, the top seller, is a below-average earner.
7. Regional performance is balanced — no underperforming region.
8. 81% of customers are repeat buyers — retention appears to be the stronger revenue driver.
9. Revenue is not concentrated in a small number of customers.
10. Furniture and Home & Kitchen offer the best balance of sales volume and margin.

Full detail with evidence: `documentation/insights.md`.

10. Recommendations

1. Cap routine promotional discounts at 20%.
2. Review Electronics pricing and cost structure.
3. Reassess USB-C Charger's place in the catalog.
4. Shift incremental marketing/inventory investment toward Furniture and Home & Kitchen.

5. Plan inventory and staffing ahead of the October–November peak.
6. Build a retention program around the existing repeat-customer base.
7. Investigate the one-time-buyer segment to understand non-return reasons.

Full detail with supporting evidence: `documentation/recommendations.md`.

11. Conclusion

The business shows healthy overall profitability (24.2% margin) on a loyal, well-distributed customer base. However, performance is uneven beneath the surface: the top category and top product by revenue are both below-average earners, and heavy discounting is actively destroying margin at the highest tiers. The recommendations in this report are aimed directly at these gaps — correcting margin erosion, rebalancing category investment, and reinforcing the retention pattern that already drives most of the business.

Dataset is synthetic and was generated for this project; all figures are internally consistent and traceable to the accompanying Excel workbook.